A Priori Error Analysis

Interpolation estimates and finite element error bounds

Lecture outline

This lecture begins where the previous lecture stopped: after the proof of Céa's lemma.

1. Construct the nodal interpolant and prove the interpolation estimates for linear elements in one dimension.
2. State the corresponding result on shape-regular triangular and tetrahedral meshes.
3. Combine interpolation with the energy best-approximation property to obtain finite element error estimates.

The optimal L^2 estimate requires an additional duality argument and is the first main topic of the next lecture.

Finite element interpolation

Construct a finite element function from the nodal values of a given function.

Finite element interpolant

To use Céa's lemma, we need a specific function $v_h \in V_h$ whose error can be estimated. We choose the nodal interpolant.

Let V_h be a Lagrange finite element space with global nodes $\boldsymbol{x}_i$ and nodal basis functions ϕ_i. For a continuous function u, define

$$\boxed{\mathcal{I}_h u = \sum_{i=1}^{N_n} u(\boldsymbol{x}_i)\phi_i.}$$

The interpolant belongs to V_h and satisfies

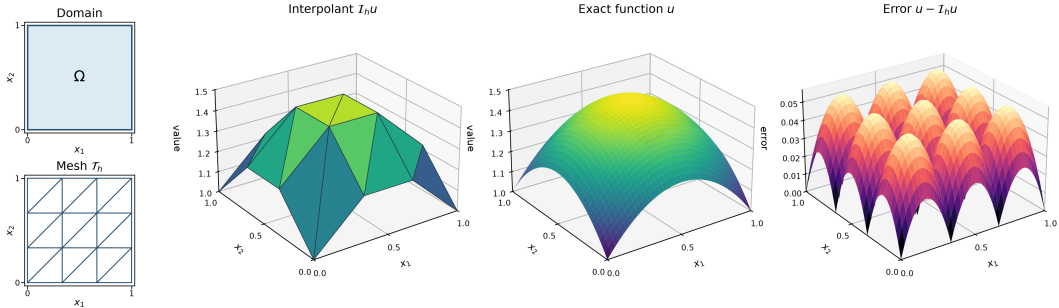
$$\mathcal{I}_h u(\boldsymbol{x}_i) = u(\boldsymbol{x}_i) \qquad i = 1, \ldots, N_n.$$

On an element $\mathcal{K}_e$,

$$(\mathcal{I}_h u)|_{\mathcal{K}_e} = \mathcal{I}_e(u|_{\mathcal{K}_e}).$$

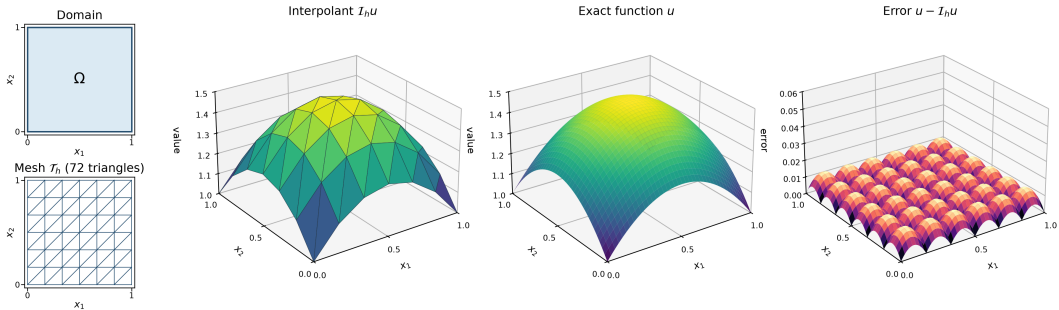
Thus the global interpolant can be constructed cell by cell from the nodal values of u.

Interpolation on a triangular mesh



The coarse mesh defines the P_1 interpolant. The finer plotting points only display the exact function and the error smoothly; they are not additional finite element nodes. For this quadratic and uniform mesh, the error pattern repeats on every triangle up to translation or reflection.

Interpolation after uniform refinement



Halving the grid spacing increases the mesh from 18 to 72 triangles. With the same vertical scales in both figures, the maximum error decreases from 1/18 to 1/72, a factor of four.

Properties and regularity of the interpolant

Three properties will be used in the error estimates.

1. **Locality:** $(\mathcal{I}_h u)|_{\mathcal{K}_e}$ depends only on the degrees of freedom of $\mathcal{K}_e$.
2. **Polynomial reproduction:** if $u|_{\mathcal{K}_e}$ belongs to the local finite element space, then $\mathcal{I}_e u = u$ on $\mathcal{K}_e$.
3. **Boundary values:** if $u = 0$ on Γ_D and the mesh resolves Γ_D, then $\mathcal{I}_h u \in \mathcal{V}_{h,0}$.

The nodal interpolant requires point values. In one dimension, H^1 functions are continuous. In two and three dimensions, a general H^1 function need not be continuous, so the estimates will assume additional regularity, commonly $u \in H^2(\Omega)$.

A four-element linear mesh

Consider $u(x) = 1 + x(1 - x)$ on $[0, 1]$. Use four uniform linear elements with five nodes:

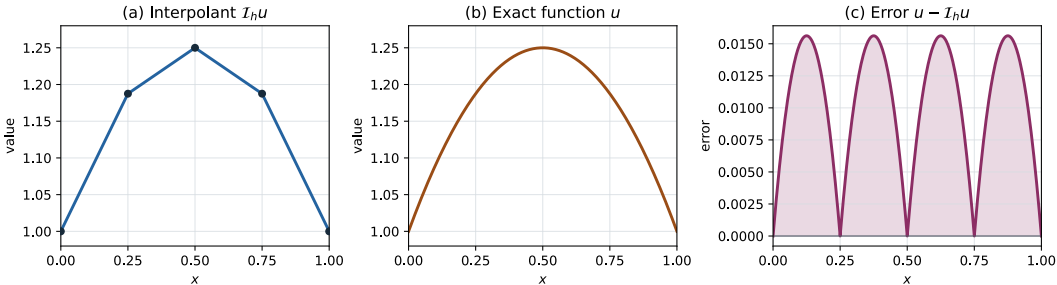
$$x_1 = 0, \qquad x_2 = \frac{1}{4}, \qquad x_3 = \frac{1}{2}, \qquad x_4 = \frac{3}{4}, \qquad x_5 = 1.$$

On $\mathcal{K}_e = [x_e, x_{e+1}]$,

$$\mathcal{I}_e u = u(x_e)\phi_1^e + u(x_{e+1})\phi_2^e.$$

The interpolant agrees with u at both element nodes and is linear between them. Adjacent local interpolants share the same value at their common node, so the global interpolant is continuous.

Exact function, interpolant, and error

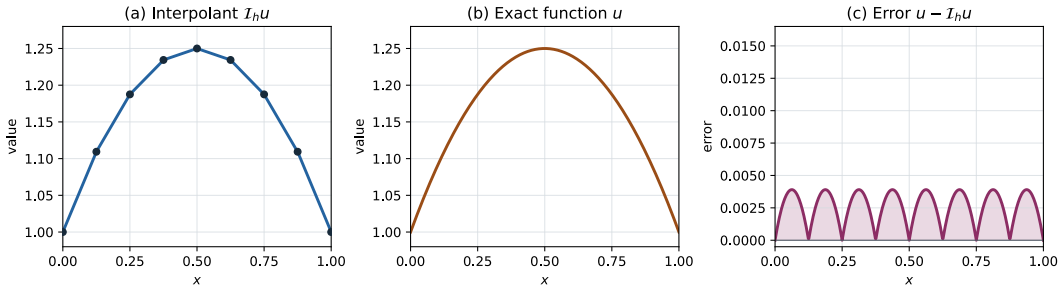


On every element,

$$u(x) - \mathcal{I}_e u(x) = (x - x_e)(x_{e+1} - x).$$

Since $h = 1/4$ on each element, the error reaches the same maximum $h^2/4 = 1/64$ at every element midpoint.

Interpolation on eight linear elements



Dividing every coarse element in two gives nine nodes $x_i = (i-1)/8$. The element length decreases from $1/4$ to $1/8$, and the maximum error decreases from $1/64$ to $1/256$.

Interpolation estimates in one dimension

Estimate the error on one element, then sum the element estimates over the mesh.

Local interpolation estimates

Let $\mathcal{K}_e = (x_e, x_{e+1})$, $h_e = x_{e+1} - x_e$, and $u \in H^2(\mathcal{K}_e)$. Then

$$\boxed{\left\|(u - \mathcal{I}_e u)'\right\|_{L^2(\mathcal{K}_e)} \leq h_e \left\|u''\right\|_{L^2(\mathcal{K}_e)}}$$

and

$$\boxed{\left\|u - \mathcal{I}_e u\right\|_{L^2(\mathcal{K}_e)} \leq h_e^2 \left\|u''\right\|_{L^2(\mathcal{K}_e)}.}$$

The derivative error contains one power of h_e. The error in the function values contains two powers of h_e.

Why the derivative error vanishes at one point

Set

$$w = u - \mathcal{I}_e u.$$

Because $\mathcal{I}_e u$ is linear, $w'' = u''$. Also,

$$w(x_e) = w(x_{e+1}) = 0.$$

Rolle's theorem. If a function is continuous on $[a, b]$, differentiable on (a, b), and has equal endpoint values, then its derivative is zero at some point in (a, b).
In one dimension, an H^2 function has a continuously differentiable representative. Rolle's theorem therefore gives

$$x_* \in (x_e, x_{e+1}) \qquad \text{such that} \qquad w'(x_*) = 0.$$

The fundamental theorem of calculus, applied to w', now gives

$$w'(y) - w'(x_*) = \int_{x_*}^{y} w''(t)\,\mathrm{d}t, \qquad w'(y) = \int_{x_*}^{y} w''(t)\,\mathrm{d}t.$$

The pointwise derivative bound

For $y \in \mathcal{K}_e$, let I_y be the interval with endpoints x_* and y. Its length satisfies $|I_y| \leq h_e$. Therefore

$$\begin{aligned} |w'(y)|^2 &= \left| \int_{I_y} 1 \, w''(t) \, \mathrm{d}t \right|^2 \\ &\leq \left(\int_{I_y} 1^2 \, \mathrm{d}t \right) \left(\int_{I_y} |w''(t)|^2 \, \mathrm{d}t \right) \\ &\leq h_e \left\| w'' \right\|^2_{L^2(\mathcal{K}_e)} . \end{aligned}$$

This is a pointwise estimate: it bounds $|w'(y)|$ for each $y \in \mathcal{K}_e$ using one element norm of w''.

The local derivative estimate

Integrate the pointwise bound with respect to y over the element:

$$\begin{aligned}\left\|w'\right\|_{L^2(\mathcal{K}_e)}^2 &= \int_{x_e}^{x_{e+1}} |w'(y)|^2 \, \mathrm{d}y \\ &\leq \int_{x_e}^{x_{e+1}} h_e \left\|w''\right\|_{L^2(\mathcal{K}_e)}^2 \, \mathrm{d}y \\ &= h_e^2 \left\|w''\right\|_{L^2(\mathcal{K}_e)}^2 .\end{aligned}$$

Since $w'' = u''$, taking square roots gives

$$\boxed{\left\|(u - \mathcal{I}_e u)'\right\|_{L^2(\mathcal{K}_e)} \leq h_e \left\|u''\right\|_{L^2(\mathcal{K}_e)} .}$$

The function-value estimate

Use $w(x_e) = 0$. The fundamental theorem of calculus gives

$$w(y) - w(x_e) = \int_{x_e}^{y} w'(t)\,\mathrm{d}t, \qquad w(y) = \int_{x_e}^{y} w'(t)\,\mathrm{d}t.$$

For each $y \in \mathcal{K}_e$, Cauchy–Schwarz gives

$$\begin{aligned} |w(y)|^2 &\leq \left(\int_{x_e}^{y} 1^2\,\mathrm{d}t\right)\left(\int_{x_e}^{y} |w'(t)|^2\,\mathrm{d}t\right) \\ &\leq h_e \left\|w'\right\|^2_{L^2(\mathcal{K}_e)}. \end{aligned}$$

Integrating over $y \in \mathcal{K}_e$ and taking square roots gives

$$\|w\|_{L^2(\mathcal{K}_e)} \leq h_e \left\|w'\right\|_{L^2(\mathcal{K}_e)}.$$

Combining this bound with the derivative estimate gives

$$\boxed{\|u - \mathcal{I}_e u\|_{L^2(\mathcal{K}_e)} \leq h_e^2 \left\|u''\right\|_{L^2(\mathcal{K}_e)}.}$$

From local estimates to global estimates

Because the elements have disjoint interiors,

$$|u - \mathcal{I}_h u|^2_{H^1(0,L)} = \sum_{e=1}^{N} \left\| (u - \mathcal{I}_e u)' \right\|^2_{L^2(\mathcal{K}_e)},$$

and

$$\|u - \mathcal{I}_h u\|^2_{L^2(0,L)} = \sum_{e=1}^{N} \|u - \mathcal{I}_e u\|^2_{L^2(\mathcal{K}_e)}.$$

Applying the local bounds gives

$$|u - \mathcal{I}_h u|^2_{H^1(0,L)} \leq \sum_{e=1}^{N} h_e^2 \left\| u'' \right\|^2_{L^2(\mathcal{K}_e)},$$

$$\|u - \mathcal{I}_h u\|^2_{L^2(0,L)} \leq \sum_{e=1}^{N} h_e^4 \left\| u'' \right\|^2_{L^2(\mathcal{K}_e)}.$$

Global interpolation estimates

Let

$$h = \max_{1 \le e \le N} h_e.$$

Since $h_e \le h$ for every element,

$$|u - \mathcal{I}_h u|^2_{H^1(0,L)} \le h^2 \sum_{e=1}^{N} \left\|u''\right\|^2_{L^2(\mathcal{K}_e)} = h^2 \left\|u''\right\|^2_{L^2(0,L)},$$

and

$$\|u - \mathcal{I}_h u\|^2_{L^2(0,L)} \le h^4 \sum_{e=1}^{N} \left\|u''\right\|^2_{L^2(\mathcal{K}_e)} = h^4 \left\|u''\right\|^2_{L^2(0,L)}.$$

Taking square roots gives

$$\boxed{|u - \mathcal{I}_h u|_{H^1(0,L)} \le h \left\|u''\right\|_{L^2(0,L)}, \qquad \|u - \mathcal{I}_h u\|_{L^2(0,L)} \le h^2 \left\|u''\right\|_{L^2(0,L)}.}$$

Full H^1-norm estimate

Let $e = u - \mathcal{I}_h u$. By the definition of the H^1 norm,

$$\|e\|^2_{H^1(0,L)} = \|e\|^2_{L^2(0,L)} + |e|^2_{H^1(0,L)}\,.$$

Using the two global interpolation estimates gives

$$\begin{aligned}\|u - \mathcal{I}_h u\|^2_{H^1(0,L)} &\le \left(h^4 + h^2\right) \left\|u''\right\|^2_{L^2(0,L)} \\ &= h^2 \left(1 + h^2\right) \left\|u''\right\|^2_{L^2(0,L)}\,.\end{aligned}$$

Therefore

$$\boxed{\|u - \mathcal{I}_h u\|_{H^1(0,L)} \le h\sqrt{1 + h^2}\, \left\|u''\right\|_{L^2(0,L)}\,.}$$

In particular, if $h \le 1$,

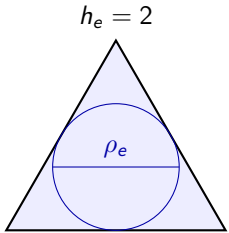
$$\|u - \mathcal{I}_h u\|_{H^1(0,L)} \le \sqrt{2}\, h \left\|u''\right\|_{L^2(0,L)}\,.$$

Thus the full H^1-norm interpolation error is also of order h.

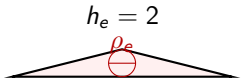
Interpolation in several dimensions

The estimate must account for both cell size and cell shape.

Cell size alone does not control interpolation



well-shaped triangle



flat triangle

Both cells have the same diameter, but the diameter ρ_e of the largest inscribed disk is much smaller for the flat cell. The next two slides show how this difference enters the affine element map and the basis-function gradients.

A family of affine triangles with the same diameter

Let the reference triangle have vertices

$$\widehat{\boldsymbol{x}}_1 = (0,0), \qquad \widehat{\boldsymbol{x}}_2 = (1,0), \qquad \widehat{\boldsymbol{x}}_3 = (0,1).$$

Map it to the physical triangle $\mathcal{K}_\varepsilon$ with vertices

$$\boldsymbol{x}_1 = (-1,0), \qquad \boldsymbol{x}_2 = (1,0), \qquad \boldsymbol{x}_3 = (0,\varepsilon),$$

using

$$\boldsymbol{x} = F_\varepsilon(\widehat{\boldsymbol{x}}) = \boldsymbol{x}_1 + \boldsymbol{J}_\varepsilon \widehat{\boldsymbol{x}}, \qquad \boldsymbol{J}_\varepsilon = \begin{bmatrix} 2 & 1 \\ 0 & \varepsilon \end{bmatrix}.$$

For $0 < \varepsilon \leq \sqrt{3}$, the longest side is the base from $\boldsymbol{x}_1$ to $\boldsymbol{x}_2$, so every triangle in this family has $h_e = 2$. Decreasing ε changes the cell shape without changing its diameter.

The inverse Jacobian amplifies a basis gradient

The inverse transpose of the Jacobian is

$$\boldsymbol{J}_{\varepsilon}^{-T} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2\varepsilon} & \frac{1}{\varepsilon} \end{bmatrix}.$$

Choose the reference P_1 basis function associated with the third vertex:

$$\widehat{v}(\xi, \eta) = \eta, \qquad \nabla_{\widehat{\boldsymbol{x}}} \widehat{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The gradient transformation from Chapter 6 gives

$$\nabla_{\boldsymbol{x}} v = \boldsymbol{J}_{\varepsilon}^{-T} \nabla_{\widehat{\boldsymbol{x}}} \widehat{v} = \begin{bmatrix} 0 \\ 1/\varepsilon \end{bmatrix}, \qquad \|\nabla_{\boldsymbol{x}} v\| = \frac{1}{\varepsilon}.$$

Thus $\left\|\boldsymbol{J}_{\varepsilon}^{-1}\right\| \geq 1/\varepsilon$. The basis gradient remains moderate for $\varepsilon = \sqrt{3}$, but has magnitude 4 when $\varepsilon = 1/4$.

The same example gives the shape parameter

The area of $\mathcal{K}_\varepsilon$ is ε, and its semiperimeter is

$$1 + \sqrt{1+\varepsilon^2}.$$

The radius of the inscribed disk is area divided by semiperimeter. Since ρ_e is the disk diameter,

$$\rho_e = \frac{2\varepsilon}{1+\sqrt{1+\varepsilon^2}}, \qquad \frac{h_e}{\rho_e} = \frac{1+\sqrt{1+\varepsilon^2}}{\varepsilon}.$$

As $\varepsilon \to 0$,

$$\left\| \boldsymbol{J}_\varepsilon^{-1} \right\| \text{ grows at least like } \frac{1}{\varepsilon}, \qquad \frac{h_e}{\rho_e} \text{ grows like } \frac{2}{\varepsilon}.$$

The inverse-Jacobian factor is therefore controlled only when the cell-shape ratio is controlled.

Shape regularity

For a cell $\mathcal{K}_e$:

- $h_e = \operatorname{diam}(\mathcal{K}_e)$ is the largest distance between two points in the cell;
- ρ_e is the diameter of the largest ball contained in the cell.

The ratio h_e/ρ_e increases as a triangle or tetrahedron becomes flatter.

A family of meshes $\{\mathcal{T}_h\}$ is **shape regular** if one constant σ satisfies

$$\boxed{\frac{h_e}{\rho_e} \leq \sigma}$$ for every cell in every mesh of the family.

Refinement may create smaller cells, but it may not create cells that become progressively flatter.

Local interpolation estimate

The estimate is first obtained on the reference simplex and transferred to $\mathcal{K}_e$ by the affine element map. Shape regularity keeps the transformation constants independent of h_e.
Let $\mathcal{I}_h u$ be the degree-p Lagrange interpolant. If

$$u \in H^s(\mathcal{K}_e), \qquad 2 \le s \le p+1,$$

then

$$\boxed{\|u - \mathcal{I}_h u\|_{L^2(\mathcal{K}_e)} \le C_I h_e^s \, |u|_{H^s(\mathcal{K}_e)}}$$

and

$$\boxed{|u - \mathcal{I}_h u|_{H^1(\mathcal{K}_e)} \le C_I h_e^{s-1} \, |u|_{H^s(\mathcal{K}_e)} .}$$

The constant C_I is independent of h_e. It may depend on the polynomial degree and on the common shape bound σ.

From local interpolation error to global error

The element interiors do not overlap, so

$$\|u - \mathcal{I}_h u\|^2_{L^2(\Omega)} = \sum_e \|u - \mathcal{I}_h u\|^2_{L^2(\mathcal{K}_e)} .$$

Using the local estimate and $h_e \leq h := \max_e h_e$,

$$\begin{aligned} \|u - \mathcal{I}_h u\|^2_{L^2(\Omega)} &\leq C_I^2 \sum_e h_e^{2s} \, |u|^2_{H^s(\mathcal{K}_e)} \\ &\leq C_I^2 h^{2s} \, |u|^2_{H^s(\Omega)} . \end{aligned}$$

Therefore

$$\|u - \mathcal{I}_h u\|_{L^2(\Omega)} \leq C_I h^s \, |u|_{H^s(\Omega)} ,$$

and the same calculation gives

$$|u - \mathcal{I}_h u|_{H^1(\Omega)} \leq C_I h^{s-1} \, |u|_{H^s(\Omega)} .$$

Key result from interpolation

For a shape-regular family of affine simplicial meshes, degree-p Lagrange interpolation, and

$$u \in H^s(\Omega), \qquad 2 \le s \le p+1,$$

the global estimates are

$$\boxed{\|u - \mathcal{I}_h u\|_{L^2(\Omega)} \le C_I h^s \, |u|_{H^s(\Omega)},}$$

$$\boxed{|u - \mathcal{I}_h u|_{H^1(\Omega)} \le C_I h^{s-1} \, |u|_{H^s(\Omega)}.}$$

For $h \le 1$, these two bounds also give

$$\|u - \mathcal{I}_h u\|_{H^1(\Omega)} \le \sqrt{2}\, C_I h^{s-1} \, |u|_{H^s(\Omega)}.$$

The constant C_I is independent of h.

Finite element error estimates

Combine Céa's best-approximation result with the interpolation estimates.

Céa's lemma supplies the comparison

The previous lecture proved Céa's estimate

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in \mathcal{V}_{h,0}} \|u - v_h\|_V .$$

For symmetric diffusion, choose the energy norm. In this norm the best-approximation constant is one, so

$$\|u - u_h\|_a \leq \|u - v_h\|_a \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

For homogeneous Dirichlet data on Γ_D, with a mesh that resolves Γ_D, the interpolant satisfies $\mathcal{I}_h u \in \mathcal{V}_{h,0}$. Set $v_h = \mathcal{I}_h u$:

$$\boxed{\|u - u_h\|_a \leq \|u - \mathcal{I}_h u\|_a .}$$

Energy-norm finite element error

The coefficient bound $k(\boldsymbol{x}) \leq k_1$ gives, for any $w \in \mathcal{V}_0$,

$$\|w\|_a^2 = \int_\Omega k|\nabla w|^2 \, \mathrm{d}\boldsymbol{x} \leq k_1 \, |w|^2_{H^1(\Omega)} \, .$$

Set $w = u - \mathcal{I}_h u$ and use the two results in sequence:

$$\begin{aligned} \|u - u_h\|_a &\leq \|u - \mathcal{I}_h u\|_a && \text{(best approximation)} \\ &\leq \sqrt{k_1} \, |u - \mathcal{I}_h u|_{H^1(\Omega)} && \text{(coefficient bound)} \\ &\leq C_I \sqrt{k_1} \, h^{s-1} \, |u|_{H^s(\Omega)} && \text{(interpolation estimate).} \end{aligned}$$

Therefore

$$\boxed{\|u - u_h\|_a \leq C_I \sqrt{k_1} \, h^{s-1} \, |u|_{H^s(\Omega)} \, .}$$

From energy error to H^1 error

For $v \in \mathcal{V}_0$, Poincaré's inequality gives

$$\|v\|^2_{H^1(\Omega)} = \|v\|^2_{L^2(\Omega)} + |v|^2_{H^1(\Omega)} \leq (1 + C_P^2)\, |v|^2_{H^1(\Omega)}\,.$$

The lower coefficient bound gives

$$\|v\|^2_a = \int_\Omega k |\nabla v|^2 \, \mathrm{d}\boldsymbol{x} \geq k_0\, |v|^2_{H^1(\Omega)}\,.$$

Consequently,

$$\|v\|_{H^1(\Omega)} \leq \sqrt{\frac{1 + C_P^2}{k_0}}\, \|v\|_a\,.$$

Set $v = u - u_h$ and substitute the energy estimate:

$$\boxed{\|u - u_h\|_{H^1(\Omega)} \leq C_I \sqrt{\frac{k_1(1 + C_P^2)}{k_0}} h^{s-1}\, |u|_{H^s(\Omega)}\,.}$$

The L^2 estimate and the remaining question

With $e = u - u_h$,

$$\|e\|_{L^2(\Omega)} \leq \|e\|_{H^1(\Omega)} \leq Ch^{s-1} |u|_{H^s(\Omega)} .$$

The argument used today therefore gives

$$\boxed{\|u - u_h\|_{L^2(\Omega)} = O(h^{s-1}).}$$

The L^2 interpolation estimate has one higher power of h:

$$\boxed{\|u - \mathcal{I}_h u\|_{L^2(\Omega)} = O(h^s).}$$

Céa's lemma controls the error in the energy norm, so it does not directly transfer this L^2 interpolation estimate to $u - u_h$.

The duality argument developed in the next lecture provides the missing power of h. We will prove

$$\boxed{\|u - u_h\|_{L^2(\Omega)} \leq Ch^s |u|_{H^s(\Omega)} .}$$